Candidate Name: ______________________________________ CT Group: 08S______

HWA CHONG INSTITUTION

C2 PRELIMINARY EXAMINATION
9746 H2 CHEMISTRY
Paper 2 Structured

25 September 2009 1 hour 30 min

Do not open this booklet until you are told to do so.

READ THESE INSTRUCTIONS FIRST

Write your name and class clearly in the spaces at the top of this page.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Answer **all** questions in the spaces provided in this question booklet.

A *Data Booklet* is provided.

You may use a calculator.

The number of marks is given in brackets [ ] at the end of each question or part question.

You are reminded of the need for good English and clear presentation in your answers.

For Examiner's Use	
1	/ 6
2	/ 13
3	/ 10
4	/ 9
5	/ 9
6	/ 13
Total	/ 60

Answer **all** questions.

1 **(a)** The alkaline earth metals (beryllium, magnesium, calcium, strontium and barium) all have an oxidation number of +2 in their compounds, and not +1 or +3.

Explain the above statement as fully as you can.

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(b) The solubility products of some Group II fluorides, at 25 °C, are given in the table below:

	Numerical values of K_{sp}
BaF_2	1.84×10^{-7}
CaF_2	3.45×10^{-11}

A student accidentally mixed 25.0 cm^3 of 0.100 mol dm^{-3} $CaCl_2$ solution with 25.0 cm^3 of 0.100 mol dm^{-3} $BaCl_2$ solution in the laboratory. To separate the two metal ions, he added just enough solid KF to precipitate the maximum amount of CaF_2 from the mixture, without precipitating BaF_2.

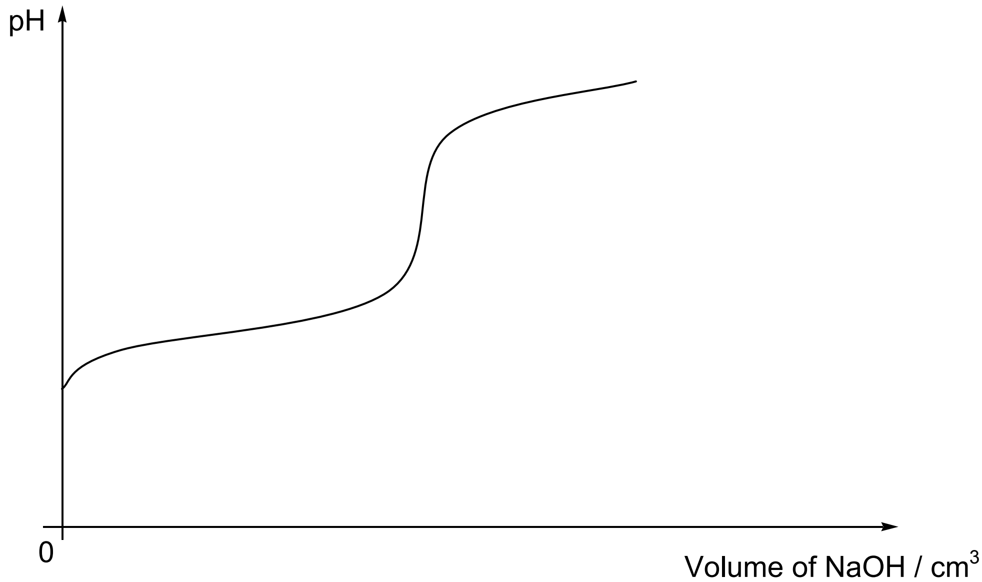
(i) Determine the $[F^-]$ required for this separation.

(ii) Determine the concentration of Ca^{2+} remaining in the final solution.

[3]
[Total: 6]

2 Glycolic acid $CH_2(OH)CO_2H$ is an α-hydroxy acid (AHA) used in various skin-care products.

(a) 0.200 g of glycolic acid (M_r = 76.0) was dissolved in 20.0 cm^3 water, and the solution was titrated with 0.100 mol dm^{-3} aqueous NaOH. The following titration curve was obtained. (K_a of glycolic acid = 1.48×10^{-4} mol dm^{-3})



(i) Calculate the volume of NaOH required to reach the equivalence point in the titration.

(ii) Calculate the pH of the solution at the equivalence point.

(iii) The table below lists the working ranges of some acid-base indicators.

Indicator	Working range	Low pH color	High pH color
Bromophenol blue	3.0 – 4.6	yellow	purple
Methyl red	4.4 – 6.2	red	yellow
Neutral red	6.8 – 8.0	red	yellow
Metacresol purple	7.6 – 9.2	yellow	purple
Thymolphthalein	9.3–10.5	colorless	blue

Suggest an indicator that can be used for this titration and give a reason for your choice.

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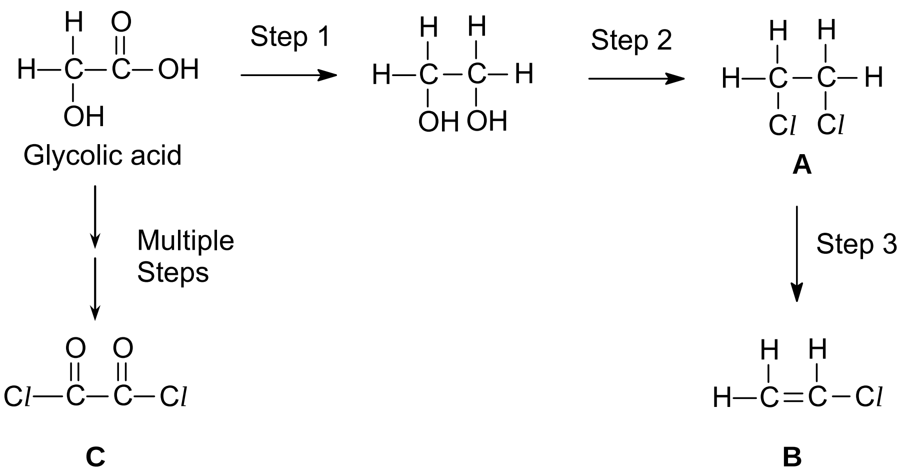
[5]

(b) Glycolic acid can undergo a series of reactions as shown below.



(i) What reagents and conditions are required for the following steps?

Step 1 …………………………………………

Step 2 …………………………………………

Step 3 …………………………………………

(ii) Arrange the three compounds **A**, **B** and **C** in order of increasing ease of hydrolysis, giving your explanations clearly.

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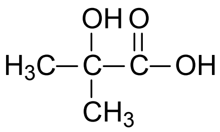
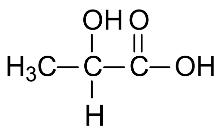
[6]

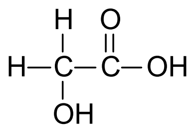
(c) Compounds **D** and **E** below are also common α-hydroxy acids.



D



E



glycolic acid

State the reagents and conditions you would use and the observations you would expect to make, in a simple chemical test, to distinguish glycolic acid from

(i) compound **D**

Reagents and conditions …………………………………………………………..

Observations ……………………………………………………………………...

(ii) compound **E**

Reagents and conditions …………………………………………………………..

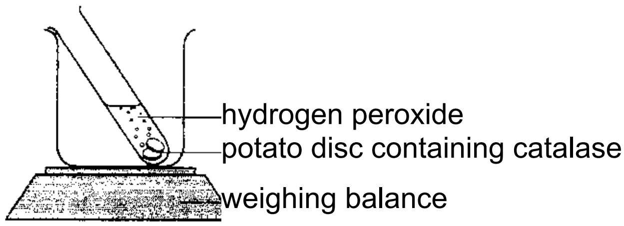
Observations ……………………………………………………………………...

[2]
[Total: 13]

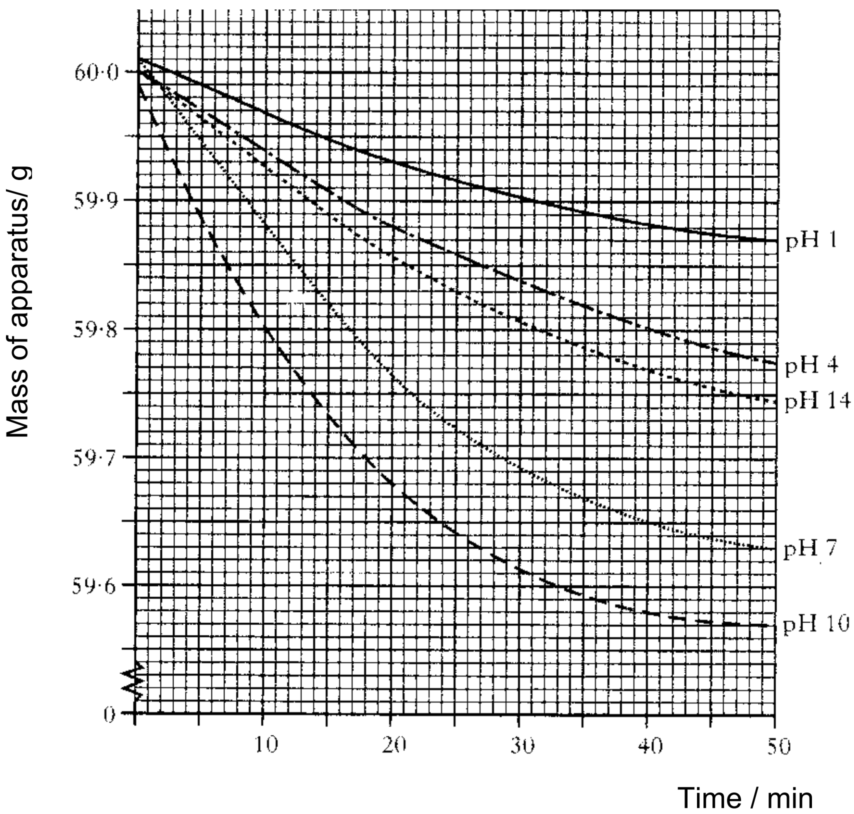
3 The decomposition of hydrogen peroxide solution to form water and oxygen gas can be catalysed by an enzyme, catalase.

$$2H_2O_2(aq) \xrightarrow{\text{catalase}} 2H_2O(l) + O_2(g)$$

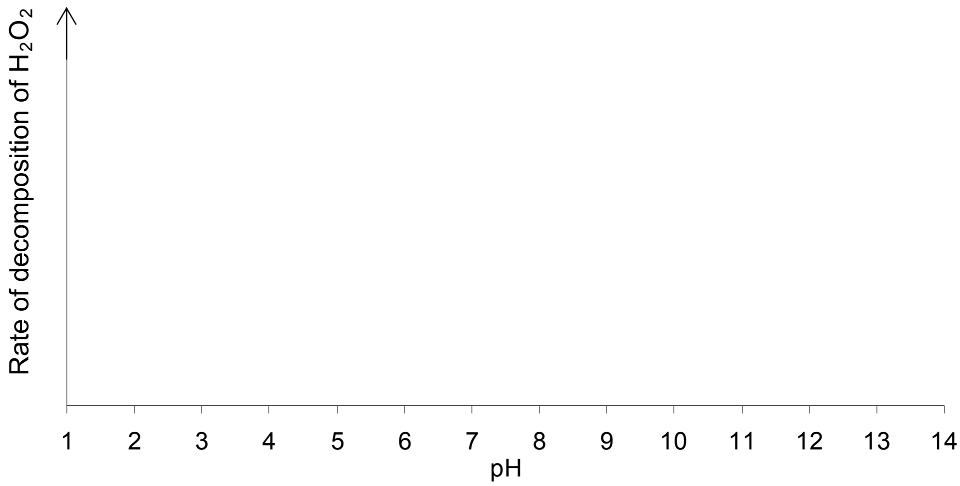
A series of experiments were carried out to study the effect of pH on the efficiency of the enzyme catalase. The apparatus was set up as shown below and its mass monitored for 50 minutes at 25°C.



The rate of reaction can be followed by recording the mass loss over a period of time. The following graphs were obtained.



(a) Using the results shown on the graphs, sketch the graph of rate of decomposition of hydrogen peroxide against pH.



[1]

(b) Explain the shape of your graph in **(a)** by considering the effect of pH on the structure of the enzyme catalase.

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(c) Explain, with the aid of a suitable diagram, why the rate of decomposition of hydrogen peroxide at pH 7 is expected to proceed faster at 35 ^oC.

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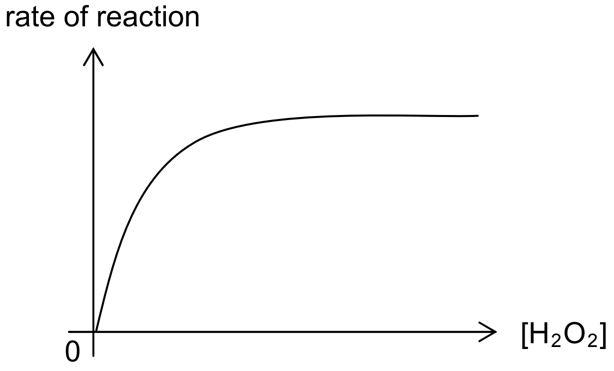
(d) Another series of experiments were carried out to study the effect of hydrogen peroxide concentration on the rate of reaction, by using different concentrations of hydrogen peroxide at constant pH 7. The following graph was obtained.



Account for the shape of the graph.

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(e) Oxygen, collected from the decomposition of hydrogen peroxide, can behave as an ideal gas under certain experimental conditions.

(i) What do you understand by the term *ideal gas*?

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(ii) Would you expect oxygen to behave as an ideal gas at a high pressure? Explain your answer.

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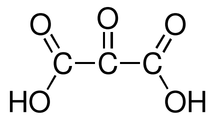
[2]
[Total: 10]

4 **(a)** 2-oxopropanedioic acid is also known as mesoxalic acid. It has been used as an antidote to cyanide poisoning.



mesoxalic acid

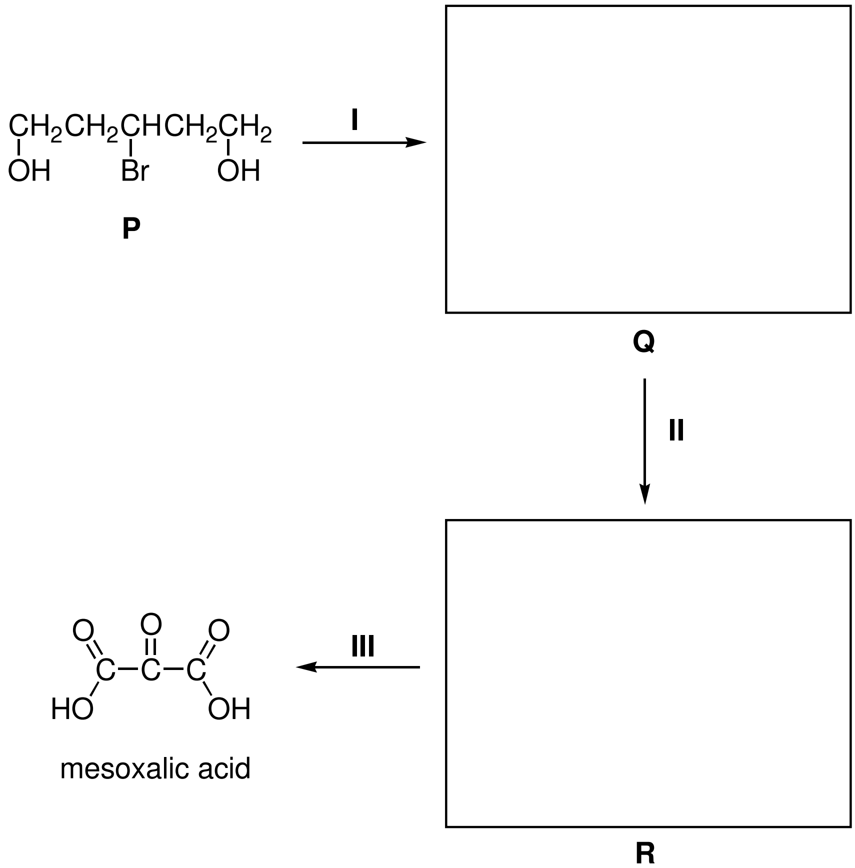
(i) When 15.0 cm^3 of 0.250 mol dm^{-3} mesoxalic acid was mixed with 20.0 cm^3 of 0.400 mol dm^{-3} sodium hydroxide, the rise in temperature was 2.8 °C. Calculate the enthalpy change of neutralization in this experiment.
(The specific heat capacity of all solutions may be taken as 4.18 J cm^{-3} K^{-1}.)

(ii) What would be the temperature rise if 10.0 cm^3 of 1.2 mol dm^{-3} sodium hydroxide had been used instead?

[4]

(b) State the reagents and conditions required for the conversion of compound **P** below into mesoxalic acid in 3 steps, and draw the structures of the intermediates

Q and **R** in the spaces provided.



Step	Reagents and conditions
I	
II	
III	

[5]
[Total: 9]

5 **(a)** Sulfuric acid, under different conditions, can behave as a dehydrating agent, a catalyst, an acid or an oxidizing agent.

(i) By means of equations, give **one** different reaction in each case where concentrated sulfuric acid behaves as a

• dehydrating agent

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• catalyst

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(ii) Using the reaction between concentrated sulfuric acid and solid sodium bromide, write an equation, including state symbols, to illustrate how concentrated sulfuric acid behaves as an

• acid

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• oxidizing agent

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[4]

(b) When chlorine gas is bubbled into a sodium halide solution, a brown coloration is observed. Upon addition of reagent **X** into the resulting mixture, a separate violet layer is observed.

(i) Suggest a possible identity of reagent **X**.

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(ii) Deduce the identity of the halide present, and write an equation for its reaction with chlorine.

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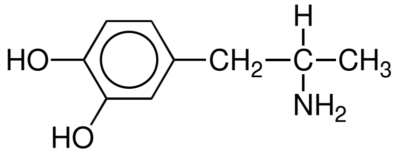
[2]

(c) Benzedrin is a stimulant and has the following structural formula:

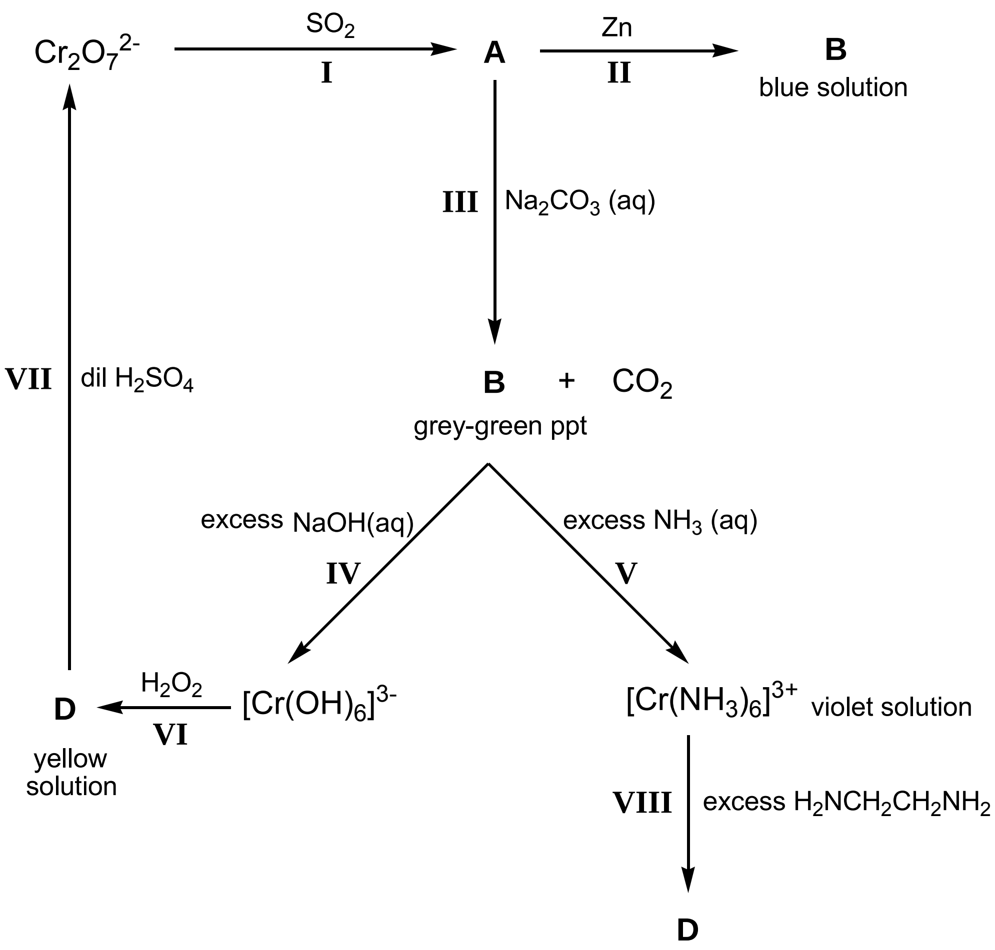


Draw the structure of the organic products when benzedrin reacts with each of the following reagents.

(i)	dilute nitric acid
(ii)	ethanoyl chloride
(iii)	NaOH (aq), followed by CH_3I and heat

[3]
[Total: 9]

6 The following reaction scheme shows the chemistry of some chromium-containing species in aqueous solution.



(a) **(i)** Write the formulae of the following chromium-containing species.

A .... **B** ....

C .... **D** ....

(ii) Draw the structure of **E** in the space below.

(iii) Explain why CO_2 is given off in reaction **III**.

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(iv) State the type of reaction that occurs in **V**.

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[7]

(b) Explain why $[Cr(NH_3)_6]^{3+}$ is coloured.

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(c) In chromium electroplating, the electrolyte is a solution containing Cr^{3+}(aq) ions. In a given electroplating experiment, a current of 3.0 A was passed through the circuit for 45 minutes to deposit a thin layer of chromium on a metal object.

(i) Calculate the mass of chromium deposited on the metal object.

(ii) Suggest why CrO_4^{2-} (aq) is usually added to the electrolyte in chromium electroplating.

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[3]
[Total: 13]